

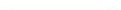
Python for Data Science II: NumPy

Write clear, testable Python and reason about data transformations.



Learning outcomes

01  Represent numerical data with arrays and meaningful shapes.

02  Replace slow loops with vectorized operations.

03  Explain broadcasting and shape mismatches.

The concept system

01

ndarrays

02

indexing

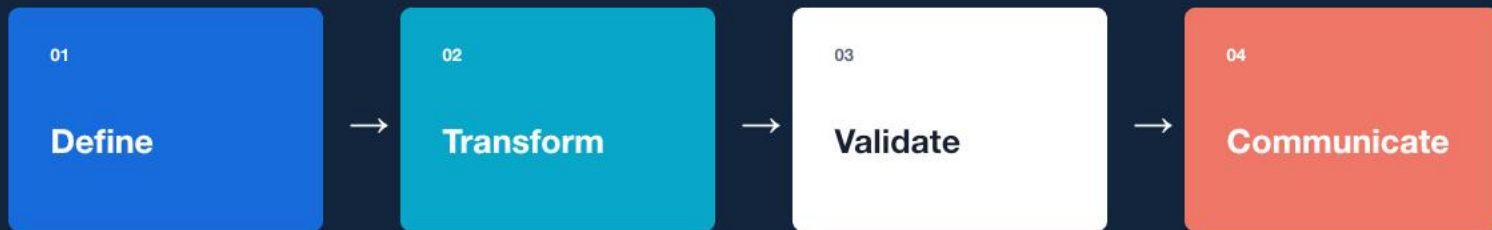
03

vectorization

04

broadcasting

A repeatable reasoning loop



Calculate cohort retention and normalized scores from matrix data.

Guided lab

LAB BRIEF

Calculate cohort retention and normalized scores from matrix data.

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A notebook comparing loop and vectorized implementations.

QUALITY GATE

- Reproducible
- Tested
- Decision-relevant

WEEKLY ASSESSMENT

Transaction Quality Profiler

Correctness 40%, clarity 25%, tests 20%, explanation 15%.

What should remain after the lesson?

CONCEPT

ndarrays + indexing

PRACTICE

Calculate cohort retention and normalized scores from matrix data.

EVIDENCE

A notebook comparing loop and vectorized implementations.
